Physics HL Extended Essay

An investigation into the relationship between the driving point for oscillation and the instantaneous center of rotation in a uniform wooden rod

Research Question: What is the relationship between the position of the driving point for oscillation and the position of the instantaneous center of rotation from the center of mass in a wooden rod, as determined through digital motion analysis?

Word count: 3976

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I. Introduction

The motivation for this investigation stems from a common observation in baseball: the "sting" felt in the hands when a baseball bat strikes a ball poorly. That is why athletes try to strike a ball on the "sweet spot," where the impact on the handle felt by the player is nearly non-existent. This phenomenon occurs because the impact is delivered at the Center of Percussion (CoP), a specific coordinate on a rigid body where the translational and rotational force generated by the impact perfectly cancel out, thus creating an instantaneous pivot that minimizes the reaction felt at the player's grip (*Center of Percussion | Harvard Natural Sciences Lecture Demonstrations*, n.d.).

As an aspiring robotics engineer, understanding these phenomena is critical beyond the realm of sports because it is an important principle in robotics. When a space robotic arm captures a target, it uses the center of percussion to minimize the impact force transmitted back to the satellite base (Paraskevas & Papadopoulos, 2013). Similarly, it is used when a robot strikes an object or makes contact with either arms or legs, impacting at the center of percussion, which minimizes or eliminates the shock forces transmitted to the robot's joints and base (Alba et al., 2010). But, a wooden rod will be used for the investigation for simplicity.

The pivot point and the CoP form a pair of conjugate points, establishing a one-to-one correspondence (Cross, 2004). Thus, this investigation seeks to move beyond the qualitative feeling of an impact to a rigorous quantitative analysis of this phenomenon. By approximating small-amplitude oscillation as a rapid succession of discrete impacts, the relationship between the driving point of oscillation and the resulting pivot will be mapped. By establishing a coordinate system centered at the Center of Mass (CM), this study aims to find the relationship between the points.

1.1 Research question

What is the relationship between the position of the driving point for oscillation and the displacement of the instantaneous center of rotation from the center of mass in a wooden rod, as determined through digital motion analysis?

II. Background knowledge

2.1 The instantaneous center of rotation (ICR)

The pivot point observed in this investigation is formally known as the instantaneous center of rotation. This stationary node forms because the applied transverse force generates translational acceleration and rotational tangential acceleration at that specific point that are equal in magnitude but opposite in direction. This causes a net initial instantaneous velocity of zero for a short period of time (Laczi et al., 2025).

2.2 The superposition of motion

The motion of a free-standing rigid rod subjected to a force can be modeled as a superposition of two distinct mechanical behaviors: translational motion and rotational motion about the center of mass (Illingworth, 1991).

2.2.1 Translational motion

According to Newton's Second Law, the linear acceleration of any point on the body is as follows (Tsokos, 2023):

$$a = \frac{F}{M}$$

2.2.2 Rotational motion

Since a free body rotates around the CM, the tangential acceleration at a specific point on the body around the CM is as follows (Tsokos, 2023):

$$\tau = F \cdot d \cdot \sin\theta$$

$$\alpha = \frac{\tau}{I}$$

$$\alpha = \frac{d \cdot F \cdot \sin\theta}{I}$$

$$a_{tan} = r \cdot \alpha = r \cdot \frac{F \cdot d}{I} = \frac{F \cdot x_{osc} \cdot x_{piv}}{I}$$

Where,

- r = Distance from the CM to a given point
- d = Distance from the CM to the point where force is applied
- x_{osc} = Position from the CM to the driving point for oscillation / cm
- x_{piv} = Position from the CM to the ICR / cm

2.3 Moment of inertia for a uniform rod

The concept is essential for understanding the motion of a body with rotational movement. For a uniform rod of length L and total mass M, the moment of inertia about its CM is given by (Tsokos, 2023):

$$I = \frac{1}{12} ML^2$$

2.4 Theoretical derivation of the relation between the position of the driving point and the ICR from the CM

The formation of an ICR/pivot requires the net velocity at the point to be zero. Since the rod is initially at rest or in small-amplitude motion, the translational and tangential accelerations determine the initial motion. The point at which the translational acceleration is perfectly balanced by the tangential acceleration is approximated as the ICR (Laczi et al., 2025).

$$a = -a_{tan}$$

Substituting the expressions of translational acceleration and tangential acceleration from above:

$$\frac{F}{M} = -\frac{F \cdot x_{osc} \cdot x_{piv}}{I}$$

$$x_{osc} \cdot x_{piv} = -\frac{I}{M}$$

$$x_{osc} \cdot x_{piv} = -\frac{\frac{1}{12} ML^2}{M}$$

$$x_{osc} \cdot x_{piv} = -\frac{L^2}{12}$$

$$\boxed{x_{piv} = -\frac{L^2}{12} x_{osc}^{-1}}$$

The positions will be treated as a two-direction plane where the direction upwards from the CM is assigned a positive value, while the direction downwards is assigned a negative value.

2.5 Theoretical implications

The derived relationship allows for three significant theoretical predictions:

1. Inverse relationship: There is an inverse mathematical relationship between the position of the driving point for oscillation and the position of the pivot point from the CM.

2. Conjugate symmetry: To allow for vector cancellation, the pivot must always manifest on the opposite side of the center of mass from the driving point for oscillation, as it creates a tangential acceleration in the opposite direction from the linear acceleration.

3. Force independence: The location of the ICR is independent of the force exerted to create an oscillation, provided the rod remains a rigid body.

2.6 Hypothesis

As the distance x_{osc} decreases, the distance x_{piv} will increase from the CM following an inverse power law, with the pivot always appearing on the opposite side of the rod's CM from the oscillating point.

2.7 Data processing and the Fourier spectrum

To identify the point of ICR with the 0 amplitude point along the rod, a periodic transverse motion was assumed to replicate a rapid succession of discrete impacts. In an ideal theoretical model, a rigid body subjected to a periodic transverse movement will exhibit Simple Harmonic Motion. The horizontal displacement of any point on the rod over time would be described by a single sinusoidal function:

$$x(t) = Asin(\omega t + \phi) + D$$

Where A is the amplitude, ω is the angular frequency, ϕ is the phase shift, and D is the equilibrium position. However, real-world experimental data collected via motion tracking is rarely a pure sine wave. The empirical displacement-time graph is a complex superposition of the driving point's movement, structural vibrations from the apparatus, and mechanical damping (Udwadia & Trifunac, 1973). Attempting to obtain a sinusoidal model created solely from the driving oscillation from data with multiple noise sources introduces significant systematic uncertainty. Thus, to extract the true harmonic motion from the driving point for oscillation, the signal must be translated from the frequency domain. This is achieved using Fourier analysis. Fourier's theorem states that any complex, periodic waveform can be mathematically decomposed into an infinite sum of simple sinusoidal waves, each with a specific frequency, amplitude, and phase (Stanford University, n.d.). Digital analysis such as the tracker converts complex signals to individual frequency components. The Fourier spectrum isolates the fundamental frequency of the rod's oscillation from noise, providing the angular frequency (ω) required to obtain the sinusoidal regression model.

III. Methodology

3.1 Justification of Method

The core methodology relies on Digital Motion Analysis using Tracker software (Brown, 2025). The software was chosen for its ability to capture the rapid, periodic motion of the rod. This approach is superior to the naked eye as it provides a quantitative and statistically rigorous method for locating the pivot. The methodology followed the process below:

1. Video acquisition and calibration: A high-frame-rate video must be recorded to minimize the effect of afterimages and get a clear video that captures the motion. The camera must be positioned parallel to the experimental setup to eliminate parallax error. Then the video is calibrated by tracking the holes in the moving rod with a point mass feature. The rod is tracked for multiple oscillations to minimize random error and acquire a sinusoidal regression for amplitude.
2. Data collection: To overcome the difficulty of visually identifying the stationary node, the horizontal movement of 19 distinct holes along the rod is tracked across several complete

oscillation cycles. With the individual displacement-time graphs obtained from each hole, the peak frequency from the Fourier spectrum was obtained to eliminate other noise present in the motion of the rod. The peak frequency that was the closest to the manually observed frequency of the driving point was set as the frequency for the sinusoidal regression. Thus, the amplitude for each hole was found.

3. Data analysis: The principle of superposition of motion states that the horizontal amplitude must vary linearly along the length of the rod, dropping to zero at the ICR. This is because while the translational acceleration stays constant throughout the wooden rod, the rotational acceleration changes linearly as the distance from the CM varies. Therefore, the measured amplitude for each hole is plotted against its vertical position from the center of mass along the rod. The Instantaneous Center of Rotation is determined as the x-intercept of the calibration line of the graph. This graphical analysis provides a quantitative measurement of the pivot's location.

3.2 Apparatus and Materials

Apparatus	**Specification**
Wooden rod	Dimensions: 100cm x 2.7cm x 2.7cm Wood type: Russian Pine Mass: 313$\pm$1g Hole position uncertainty: $\pm$0.3cm (halved diameter rounded to one decimal place)
Smartphone camera	Model: SM-S911N Brand: Samsung Camera app: Camera (Ver 16.5.02.37) Frame rate: 60 fps Resolution: 50M resolutions
Stopwatch	$\pm$0.01s
Support structure	Wood Type: Russian Pine 4 (110cm x 2.7cm x 2.7cm) 2 (30cm x 6.8cm x 2.8cm) 2 (50cm x 2.7cm x 2.7cm) 2 (3cm x 2.7cm x 2.7cm)

Measuring ruler	Uncertainty: ±0.05cm
Linear actuator	Company: Lanxstar Model: Mini telescopic linear actuator Frequency: 0.5~4.5Hz Voltage: 3~24V
Metal rod	Length: 14.00±0.05cm Diameter: 0.40±0.05cm

Table 1: Apparatus and Materials

3.3 Experimental Setup

The experimental setup is designed to allow for controlled, two-dimensional, free oscillation of the rod. Each component of the setup was designed with careful measurements.

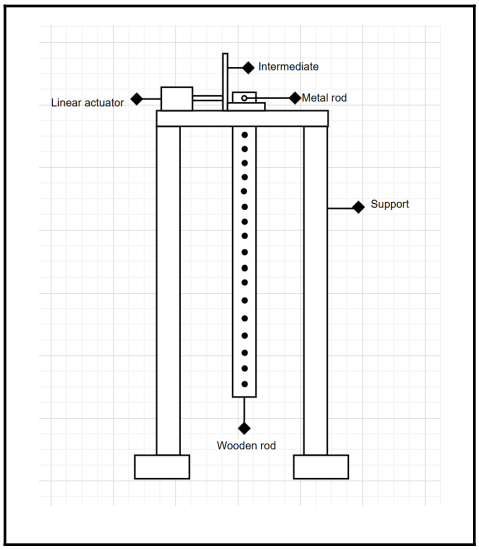


Figure 1: The experimental setup

1. Wooden rod



2.7cm x 2.7cm x 100cm

Mass: 313±1g

Figure 2: Wooden rod

A uniform wooden rod with 19 holes at a constant distance (5cm) from each other at the center of the hole was made for two reasons

1. The holes allow for a convenient change in the driving point oscillation by simply repositioning the metal rod that goes through them.
2. To track the horizontal position of the rod, a distinct mark must be visible. A standard rod without any modifications will bring trouble in tracking its horizontal movement as there is no standard point that can be tracked. The holes allow a distinct visual identification when tracking its horizontal movement with the program.

2. The support

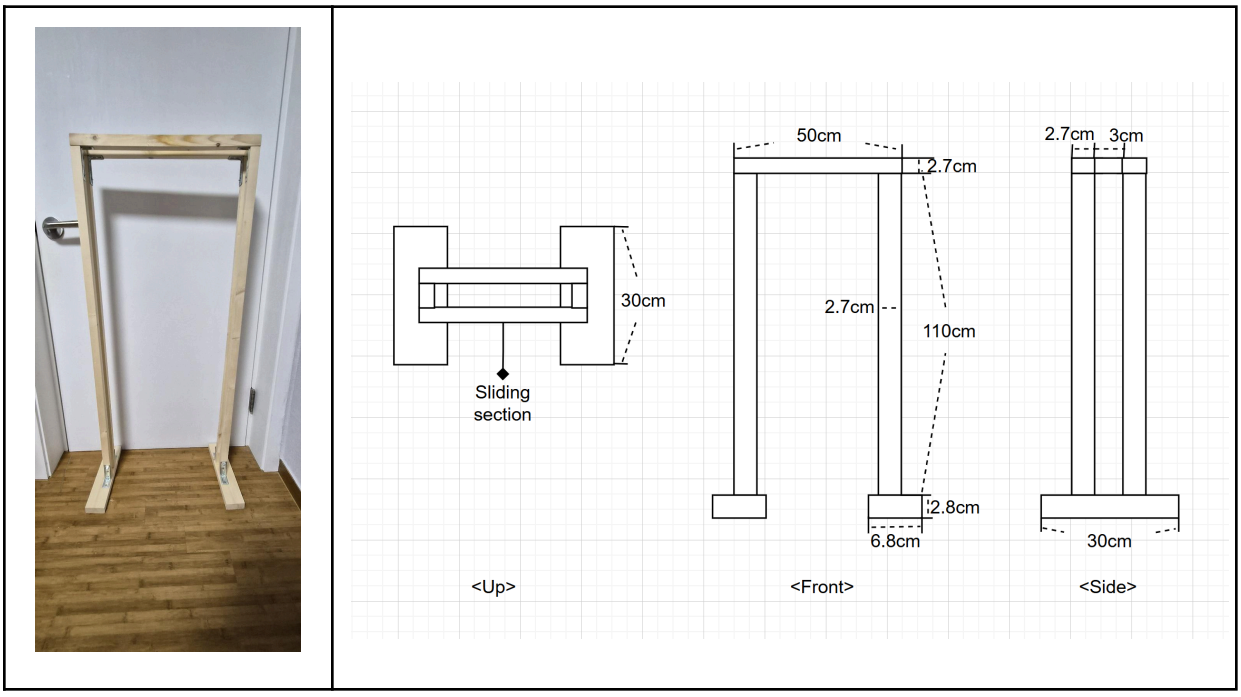


Figure 3: Support

3. Linear actuator

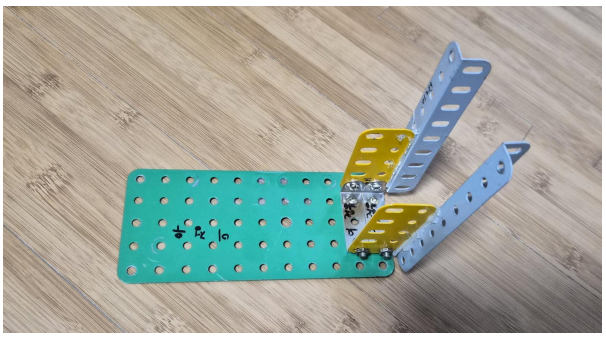


Company: Lanxstar
Model: Mini telescopic linear actuator
Frequency: 0.5~4.5Hz
Voltage: 3~24V

Figure 4: Linear actuator

A motor that allows for periodic motion along the horizontal axis with a controlled frequency was prepared to ensure identical motion for every driving point. In a preliminary experiment, it was revealed that oscillating the rod with hands manually, even with a metronome, caused significant systematic error, as humans can't create an identical motion every cycle. Therefore, to make the experiment systematic, a linear actuator was prepared. This allows isolating the relationship between the driving point and the ICR point from the frequency.

4. Intermediate



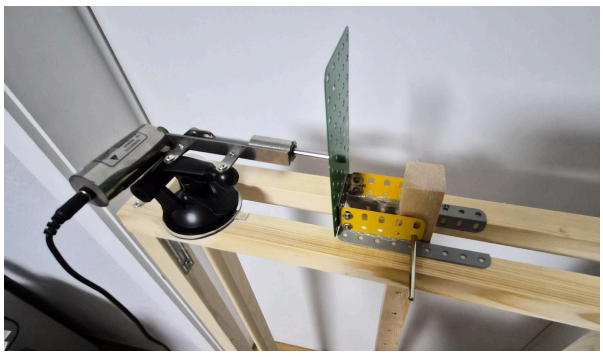


Figure 5: Intermediate/Connected with the linear actuator

In order to connect the motion of the linear actuator and the driving point along the rod, an intermediate component was required. The component was designed so that it and the rod could

slide above the support without falling off. This ensured a clean slide across the support and allowed the rod to have a horizontal periodic motion from the linear actuator.

3.4 Variables

Independent Variable (IV): The position of the driving point of oscillation (x_{osc}/cm) from the center of mass, denoted as a positive value (5.0, 10.0, 15.0, 20.0, 25.0, 30.0, 35.0, 40.0, 45.0$\pm$ 0.3cm).

- Justification of range: A 5cm interval was selected from the center of mass to ensure a maximum number of driving points along the rod without causing deterioration of the wooden rod between the holes. Moreover, the multiple ranges allow for a precise calibration for identifying the ICR.
- Method of control: The wooden rod was drilled with 19 distinct holes with a 5cm interval between them, starting from 5cm away from the end of the rod. Using a ruler ($\pm$0.05cm) to ensure a constant interval, the points were marked with a pen. Then the marked points were drilled.

Dependent Variable (DV): The position of the instantaneous center of rotation (x_{piv}/$\pm$2cm).

- Measurement method: Using tracker software, the horizontal displacement of every hole along the rod was plotted against time for each position of the driving point of oscillation. Then the amplitude of the horizontal periodic motion for each hole was found by applying a sinusoidal regression with a fixed angular frequency obtained from the Fourier spectrum to eliminate other noise. Then the measured amplitudes were plotted against the position of each hole for calibrating linear regression. Thus, the ICR was found by finding the position along the rod with 0 amplitude (x-intercept).

Controlled Variables:

Controlled Variable	**Explanation**	**Method of control**
Sliding kinetic friction	High static friction of the sliding section induces vertical bouncing and tilting for the intermediate, introducing unquantified out-of-plane motion for tracking.	Vaseline was applied to the sliding section of the support to minimize the kinetic friction coefficient.

Amplitude of Oscillation	Large amplitude oscillations create a pendulum-like arc, introducing vertical displacement and non-linear mechanical behaviour, which doesn't replicate the rapid impact on the rod.	The linear actuator's driving amplitude was constrained to a constant 1.5cm.
Frequency of Oscillation	The preliminary experiment visually revealed that the frequency of the linear actuator dictates the position of the pivot point along the rod.	The analog motor dial of the linear actuator lacked discrete markers. Thus, frequency was actively monitored and maintained within a 2.6 Hz-3.4 Hz bandwidth.
Position of the camera	Non-perpendicular viewing angles introduce parallax error, optically distorting horizontal displacement measurements in Tracker.	The camera was fixed to a stand and aligned parallel to the rod's plane of motion.

Table 2: Controlled variables

3.5 Risk Assessment

Safety considerations: During the experiment, any fragile objects must be placed away from the setup as an accidental collision with the oscillating rod could cause breakage. The experimental setup must be secured to the floor to prevent tipping that could cause a hazard to the surroundings.

Ethical: All data collection, analysis, and reporting will be conducted honestly and without bias, maintaining academic integrity.

Environmental: The materials, such as the wooden rod, support structure, and screws, will be maintained in good condition to be reusable. Any waste from the experiment will be collected and disposed of in an environmentally responsible manner.

3.6 Procedure



Figure 6: The actual experimental setup

To initiate the data collection, a metal rod was inserted through the drilled hole located 45± 0.3cm from the CM to establish a fixed driving point for oscillation. The metal rod was then connected to a linear actuator via an intermediate linkage and mounted onto the sliding section of the support base. A smartphone camera was placed on a tripod to record its continuous oscillation for 30 seconds. This process was repeated for all other driving points (5.0, 10.0, 15.0, 20.0, 25.0, 30.0, 35.0, 40.0 ±0.3cm), relocating the metal rod to each hole along one half of the wooden rod.

IV. Data

4.1 Qualitative Data

The pivot point from the naked eye moved away from the center of mass as the driving point of oscillation was getting closer to the center of mass.

4.2 Quantitative Data

4.2.1 Raw Data

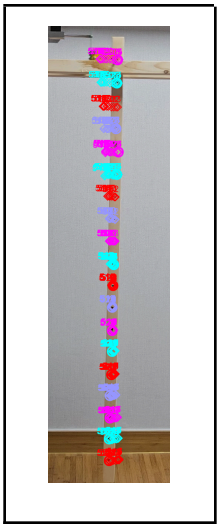


Figure 7: Tracking the holes for driving point 45 away from the CM along the rod using the tracker program

Each hole was tracked as in the picture above using the auto tracker system. The horizontal position-time graph for a position -45cm away from the CM for a driving position 45cm away from the CM was shown below:

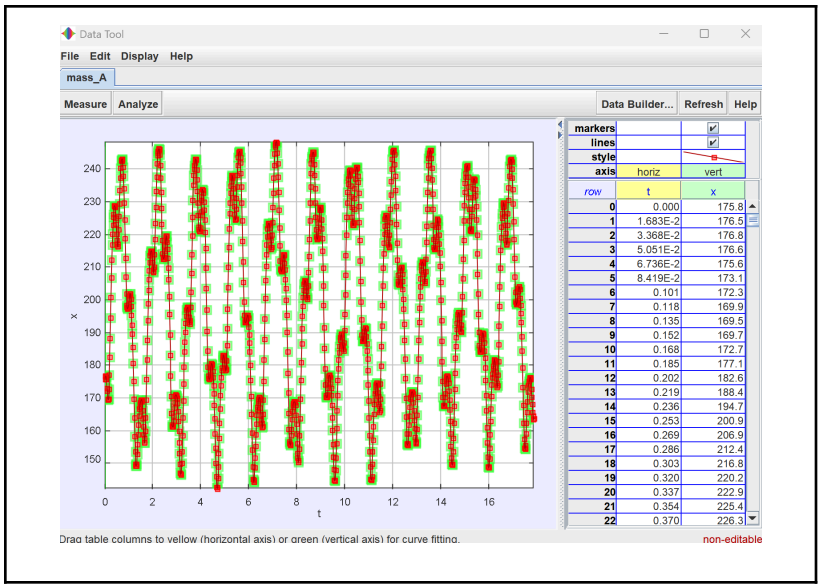
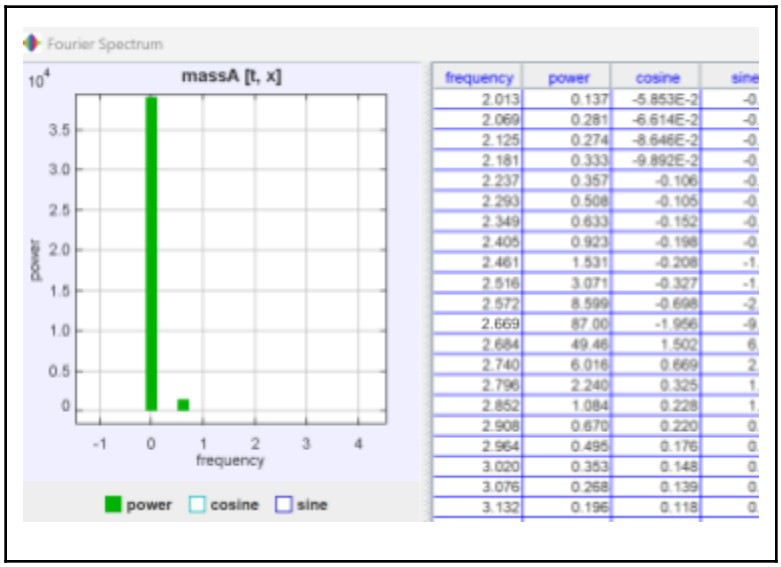


Figure 8: The result of tracking the hole -45cm for driving point 45cm from the CM with the tracker plotting the horizontal displacement with time.

The horizontal position of the point was extracted on a horizontal position-time graph. However, the graph itself is not a clean sinusoidal model with a constant amplitude that we could obtain. This indicates the presence of unavoidable noise from other external factors, likely originating from structural vibrations within the experimental setup. To isolate the rod's oscillation caused by the linear actuator, the Fourier spectrum of the graph was analyzed to understand the frequencies present in the motion.

4.2.2 Processed Data



frequency	power	cosine	sine
2.013	0.137	-5.853E-2	-0.
2.069	0.281	-6.614E-2	-0.
2.125	0.274	-8.646E-2	-0.
2.181	0.333	-9.892E-2	-0.
2.237	0.357	-0.106	-0.
2.293	0.508	-0.105	-0.
2.349	0.633	-0.152	-0.
2.405	0.923	-0.198	-0.
2.461	1.531	-0.208	-1.
2.516	3.071	-0.327	-1.
2.572	8.599	-0.698	-2.
2.669	87.00	-1.956	-9.
2.684	49.46	1.502	6.
2.740	6.016	0.669	2.
2.796	2.240	0.325	1.
2.852	1.084	0.228	1.
2.908	0.670	0.220	0.
2.964	0.495	0.176	0.
3.020	0.353	0.148	0.
3.076	0.268	0.139	0.
3.132	0.196	0.118	0.

Figure 9: The Fourier spectrum for a point -45cm away from the CM for a driving point 45cm away from the CM

The Fourier spectrum presents us with frequencies that dominate the motion of the individual rods. The highest power of the frequencies indicates that the motion is governed by that frequency the most. The 0 Hz peak represents the DC offset, which is the static average coordinate of the tracked point relative to the software's axis, and was excluded because it contains no frequency data relevant to the linear actuator (Libretexts, 2024). To determine the fundamental frequency of the linear actuator, a cross-examination was performed to identify the peak frequency closest to the experimentally measured frequency.

The time it took to complete 10 cycles was measured to be 3.38 ± 0.5 seconds. The uncertainty was set to be 0.5 seconds to account for my reaction speed. Therefore, the frequency range of the linear actuator for the driving point 45cm from the CM was calculated to be as follows:

Time taken to complete 10 cycles: 3.4±0.5 s (2s.f.)	
Calculations	**Error propagation**
Period: $\frac{Time\ taken\ to\ complete\ 10\ cycles}{10} = \frac{3.38}{10} = 0.338$ Frequency = $\frac{1}{period} = \frac{1}{0.338} = 2.95858$ $= 3.0\ (rounded\ to\ 2\ s.f.)$	Period: 0.338 Absolute uncertainty of period: $\frac{0.5}{10} = 0.05$ Relative uncertainty of period $= \frac{0.05}{0.338}$ Absolute uncertainty of frequency $= frequency \times relative\ uncertainty$ $= 2.95858 \times \frac{0.05}{0.338} = 0.43766$ $= 0.4\ (rounded\ to\ 1s.f.)$
Frequency $= 3.0 \pm 0.4$ Hz	

Table 3: Calculating the range of actual frequency generated by the linear actuator

Therefore, the peak frequency within the range of 2.6 to 3.4 was selected as the fundamental frequency caused by the linear actuator. From Figure 9, it was evident that the frequency of 2.669 was the highest with the power 87. With the same method, the peak frequency of every hole was found. The frequency for all holes with the same driving point was fairly constant (see Appendix 1 for full dataset). However, there were some deviations. In order to prevent them from drastically changing the average frequency of the rod, the outliers were detected by calculating the standard deviation.

$$Average\ frequency \pm standard\ deviation = 2.67 \pm 0.04$$

Therefore, the frequencies that were not included in the standard deviation range were excluded, and the new average frequency was calculated to be 2.67. The obtained average frequency that governs the fundamental oscillating motion from the linear actuator was required to be set as the angular frequency for the sinusoidal regression. So, the angular frequency was calculated using $\omega = 2\pi f = 2\pi \times 2.6698 = 16.77$.

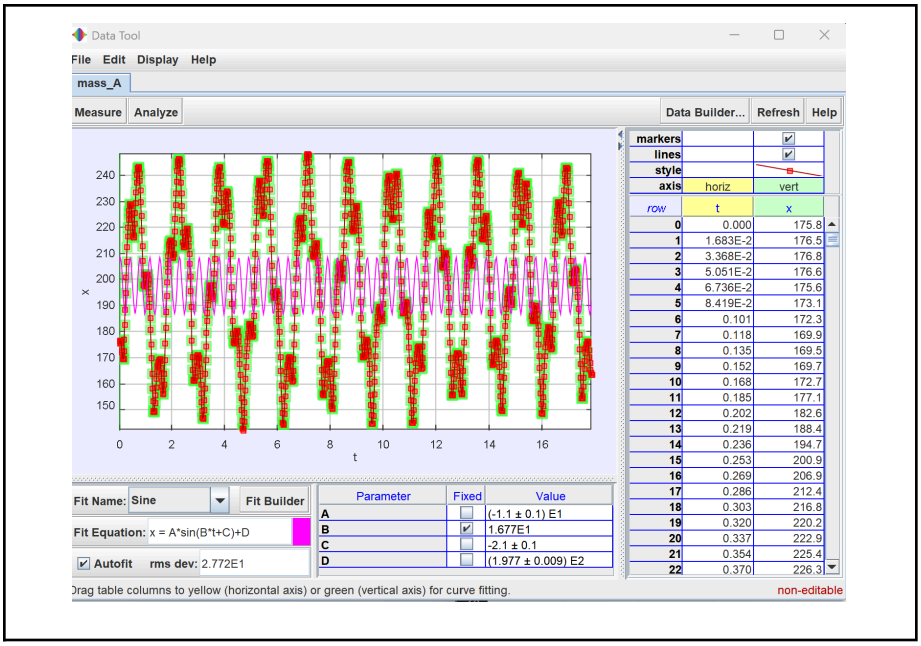


Figure 10: Applying the angular frequency to determine sinusoidal model

The sinusoidal regression has the general form of $x = A\,sin(Bt + C) + D$. The B is represented as the angular frequency of the sinusoidal model. Therefore, by plugging in the angular frequency, we obtain the amplitude of -11 ± 1. While the software outputs amplitude as a negative value, points on opposite sides of the stationary pivot oscillate 180° out of phase. Therefore, mathematical signs were assigned to these respective regions to allow a continuous linear regression across the rod. The amplitude and its absolute uncertainty for every point for the driving point 45cm away from the CM were organized as below:

Position from the center of mass (x/cm) $\pm$0.3cm	**Amplitude**	**Absolute uncertainty**
45.0	21.4	$\pm$ 0.4
40.0	20.3	$\pm$ 0.4
35.0	14.2	$\pm$ 0.7
30.0	15.1	$\pm$ 0.4
25.0	13.9	$\pm$ 0.4
20.0	12.4	$\pm$ 0.4
15.0	11.1	$\pm$ 0.8
10.0	8.4	$\pm$ 0.7
5.0	8.4	$\pm$ 0.8
0.0	4.9	$\pm$ 0.7
-5.0	3.0	$\pm$ 0.7
-10.0	1.4	$\pm$ 0.8
-15.0	-0.8	$\pm$ 0.9
-20.0	-2.4	$\pm$ 1.0
-25.0	-4.0	$\pm$ 1.0
-30.0	-6.0	$\pm$ 1.0
-35.0	-8.0	$\pm$ 1.0
-40.0	-10.0	$\pm$ 1.0
-45.0	-11.0	$\pm$ 1.0

Table 4: Data table for the extracted amplitude for each position along the wooden rod

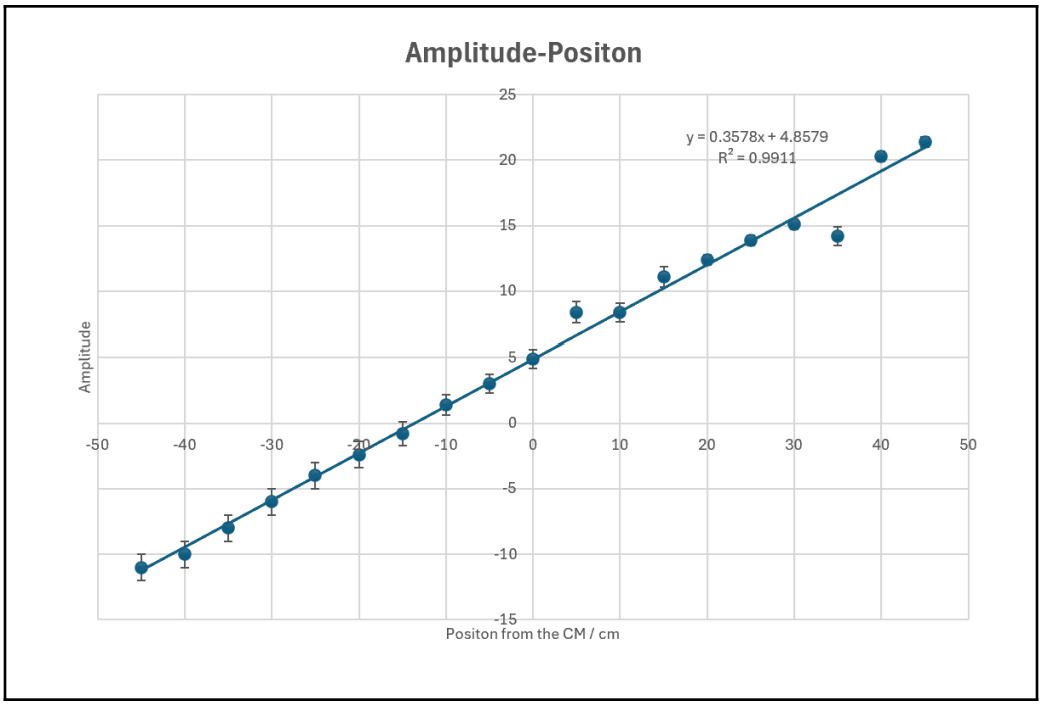


Figure 11: Amplitude plotted against the position of the hole along the wooden rod for a driving point 45cm from the CM

The horizontal uncertainty was unnoticeable. The graph indicates some outliers. Removing the outliers that are outside the regression line, we get an improved graph.

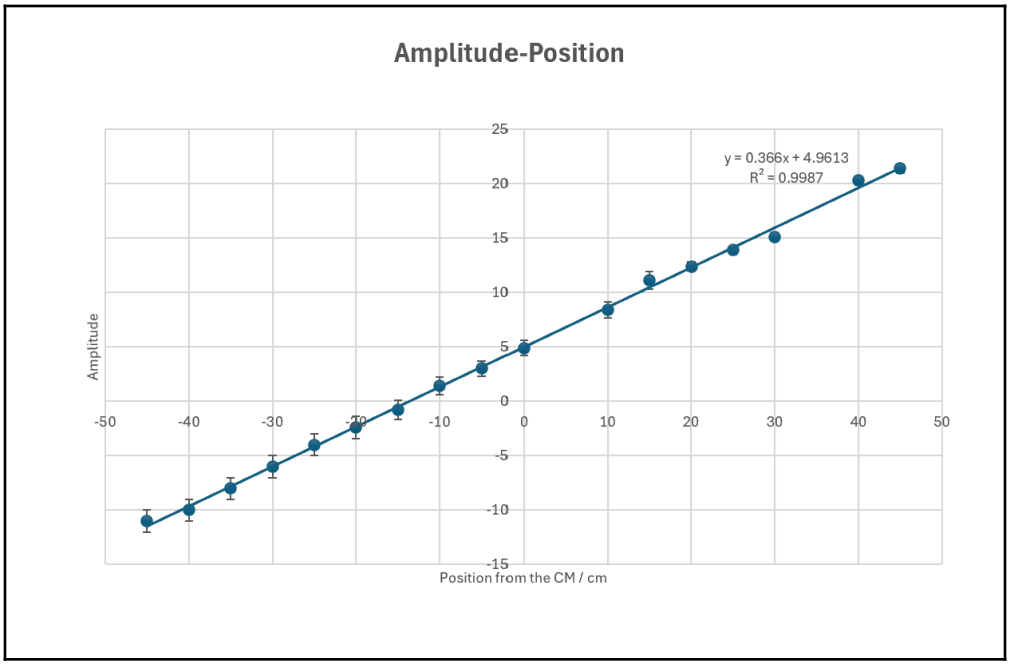


Figure 12: Refined calibration excluding the outliers

Therefore, the position with the 0 amplitude is found by analyzing the x-intercept with the best-fit line $y = 0.366x - 4.9613$ with a high R^2 value of 0.9987.

$$x\text{-intercept: } 0 = 0.366x - 4.9613$$
$$4.9613 = 0.366x$$
$$\frac{4.9613}{0.366} = 13.555 \approx 13.6$$

The maximum and minimum gradients of the calibration curve were calculated to account for the absolute uncertainty of the x-intercept. For the maximum gradient, the two extreme points were selected, and the left point had coordinates of $(x + \Delta x, y - \Delta y)$, the right point had ($x - \Delta x, y + \Delta y$). The minimum gradient was calculated where the left point was $(x - \Delta x, y + \Delta y)$ while the right point was $(x + \Delta x, y - \Delta y)$.

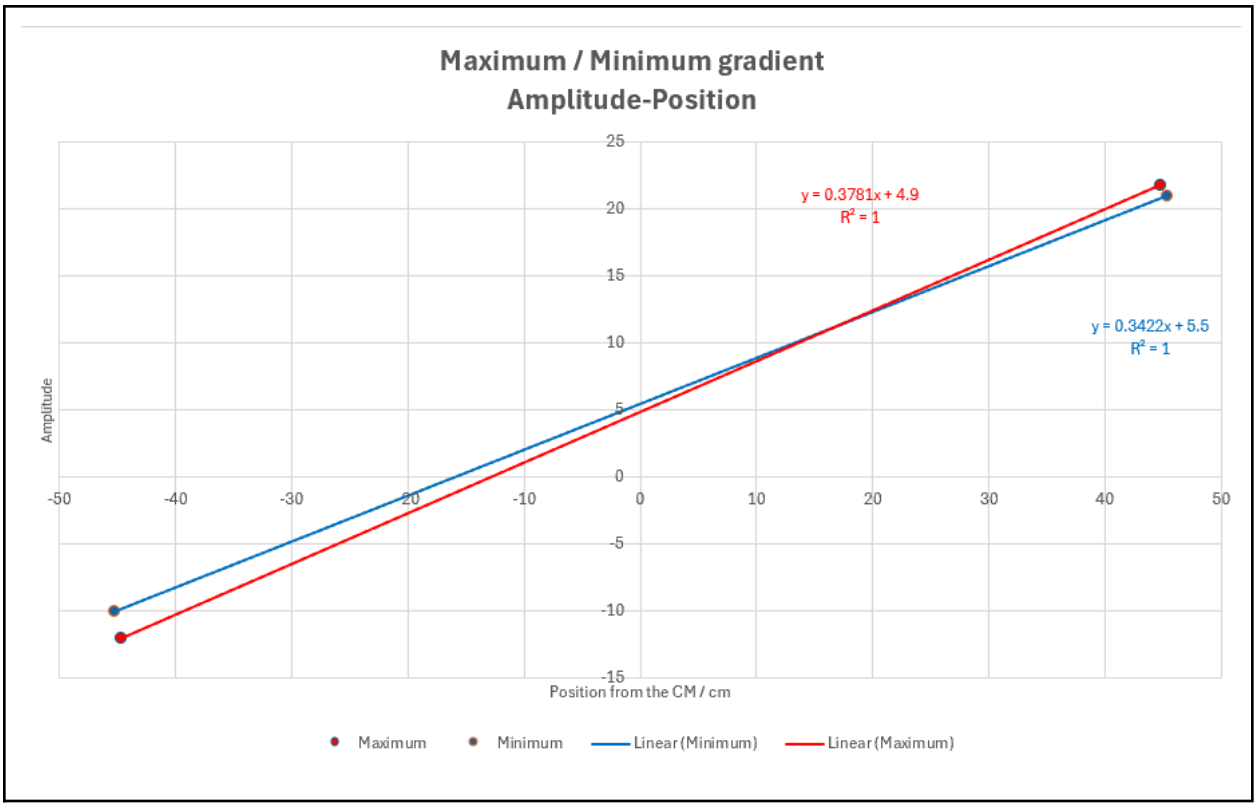


Figure 13: The maximum and minimum gradient of the calibration curve

Thus, the x-intercept for the maximum gradient is

$$y = 0.378x + 4.90$$
$$x = -\frac{4.90}{0.378} = -13.0$$

For the minimum gradient

$$y = 0.3422x + 5.50$$
$$x = -\frac{5.50}{0.342} = -16.1$$

Thus, the absolute uncertainty for the x-intercept is:

$$\frac{|(-13.0)-(-16.1)|}{2} = 1.556$$

The ICR for the driving point at 45cm from the CM is -13.555 ± 1.556. Rounding the absolute uncertainty to 1 significant figure: -14 ± 2. After determining the ICR against every driving point, the oscillation was processed to be:

Position of driving point from the CM (x_{osc}/ $\pm$0.3cm)	**Position of ICR from the CM (x_{piv}/cm)**	**Absolute uncertainty**
45.0	-14	± 2
40.0	-17	± 1
35.0	-20	± 2
30.0	-24	± 1
25.0	-29	± 2
20.0	-36	± 1
15.0	-50	± 2

10.0	-78	± 3
5.0	-155	± 10

Table 5: Processed data of the position of the driving point of oscillation and the corresponding position of ICR from the CM determined from calibration

The negative signs were turned back to positive since they only indicate directions, and the power regression required all data points to be positive. Plotting this on a graph, we get:

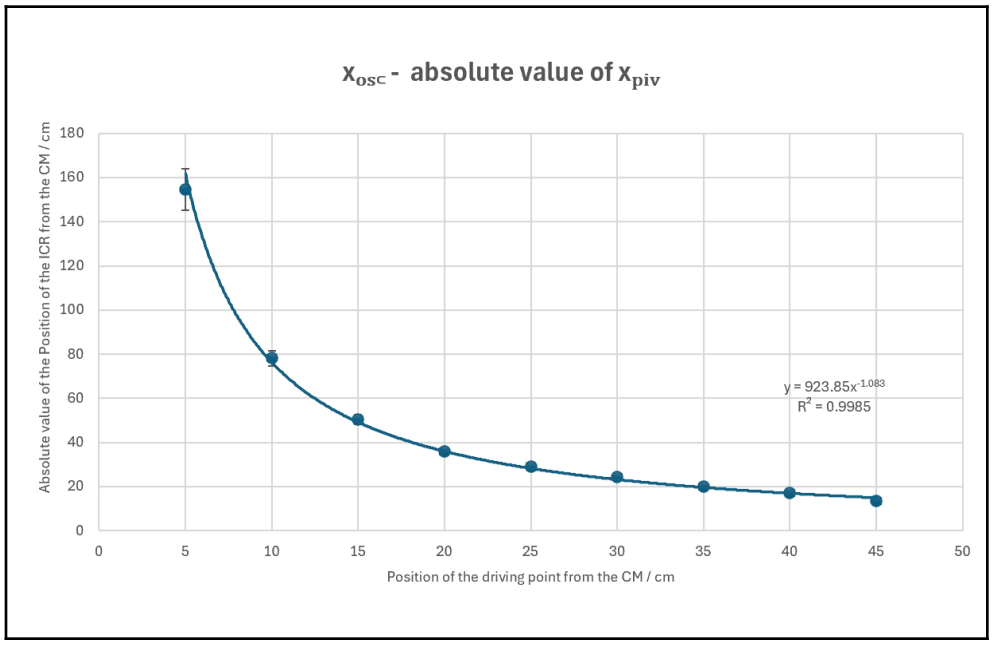


Figure 14: The absolute value of the position of the ICR plotted against the position of the driving point of oscillation from the CM

The absolute uncertainty for horizontal uncertainty was indistinguishable, and the vertical uncertainty was noticeable for only the first data point. The R^2 had a high value of 0.9983. Accounting for the direction, the relationship between the position of the driving point and the position of the ICR is revealed to be

$$x_{piv} = -923.85 x_{osc}^{-1.083}$$

The absolute uncertainty for the exponent and the constant was determined by linearization with the natural logarithm ($ln\, x_{piv} = ln\,|-923.85| - 1.083 ln\, x_{osc}$). Then, the maximum and minimum gradient of the line were determined using two extreme points' natural logarithm and their uncertainty ($\frac{\Delta value}{value}$).

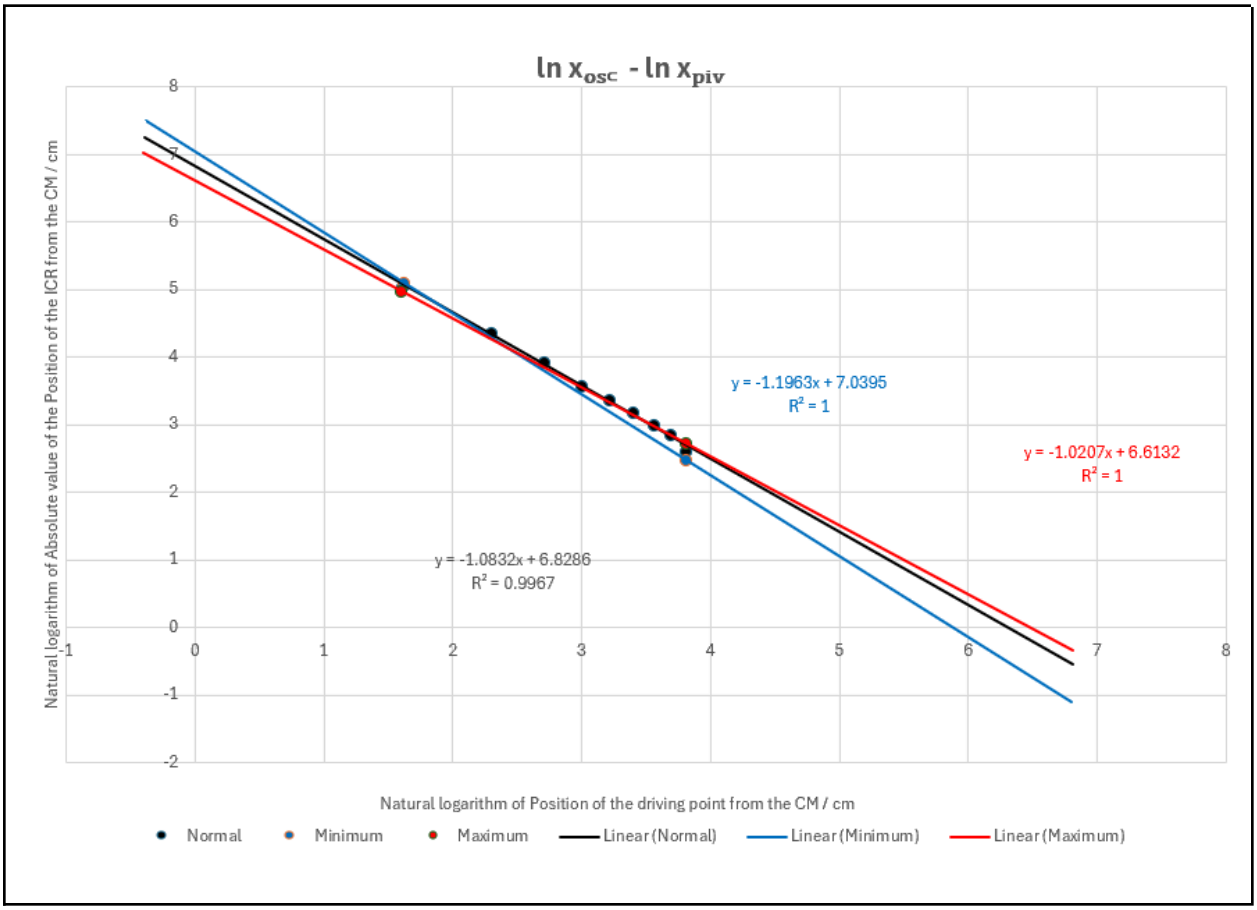


Figure 15: Maximum/minimum gradients for the linearized equation

The maximum gradient was -1.0207 while the minimum one showed -1.1963. Thus, the absolute uncertainty of the exponent -1.083 is:

$$\frac{|(-1.0207)-(-1.1963)|}{2} = 0.0878 \approx 0.09$$

With the maximum-minimum regressions, the absolute uncertainty for the constant -923.85 can be obtained with the y-intercept, which is the natural logarithm of the value. According to the

maximum gradient, the constant is -744.86 ($-e^{6.6132}$), the minimum gradient showed -1140.82 ($-e^{7.0395}$). The absolute uncertainty is:

$$\frac{|(-744.86)-(-1140.82)|}{2} = 197.98 \approx 198$$

Therefore, the final equation is $x_{piv} = (-\ 924\ \pm\ 198)x_{osc}^{-1.08\pm0.09}$

V. Conclusion

The answer to the research question "What is the relationship between the position of the driving point for oscillation and the position of the instantaneous center of rotation from the center of mass in a wooden, as determined through digital video analysis?" was found to be $x_{piv} = (-\ 924\ \pm\ 198)x_{osc}^{-1.08\pm0.09}$. This proved some theoretical implications. First, the two positions were inversely proportional, as evident from the exponent value of $-\ 1.08\ \pm\ 0.09$. Second, they had a conjugate symmetry. From Table5, we can see that for every driving point, the x-intercept (ICR) always appeared on the opposite side, indicating that the ICR always appears on the opposite side from the driving position. However, the force independence was not identified through the experiment. Nonetheless, the hypothesis " As the distance x_{osc}decreases, the distance x_{piv} will increase from the CM following an inverse power law, with the pivot always appearing on the opposite side of the rod's CM from the oscillating point." was validated from the rigorous methodologies and findings.

The empirical relationship is a direct quantitative confirmation of the superposition principle applied to rigid body motion. The formation of this ICR relies on the precise cancellation of the rod's linear translational acceleration and tangential rotational acceleration at that specific point, such that the initial net horizontal velocity is zero. The mathematical model and experimental results also reveal that the ICR can exist extending completely beyond the physical boundary of the wooden rod.

The initial motivation for this study was the concept of the sweet spot in sports equipment, formalized as the CoP. The CoP is defined as the conjugate point to a given pivot, where an impact produces zero reaction force at the pivot point. In the context of this dynamic

investigation, ICR is the pivot point of minimum reactive force from driving force on the CoP. The mapping of the qualitative relationship, $x_{piv} = (-\ 924\ \pm\ 198)x_{osc}^{-1.08\pm0.09}$, provides a precise tool for engineers and athletes. For example, to find the CoP relative to a player's grip point, one merely needs to input the grip location into the empirically found model. This finding directly applies to optimizing the design of baseball bats and tennis rackets. It also contributes to the engineering of specialized robotic manipulation by allowing designers to minimize unwanted reaction forces and structural vibration. This investigation successfully transformed a qualitative physical intuition into a precise, rigorous model.

VI. Evaluation

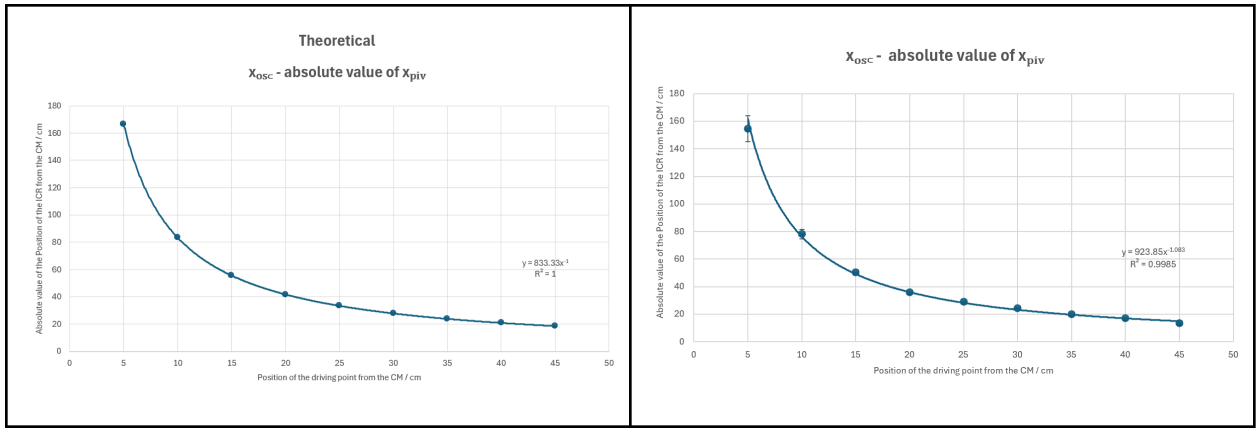


Figure 16: Comparison between the theoretical relationship and the experimental relationship

The theoretical model and the experimental model hold significant similarities, as evident from Figure16. Their equation are also identical, as the experimental exponent and the constant cover the theoretical value with their uncertainties
$(-\ 1.17 \leq -\ 1 \leq -\ 0.99, -\ 1122 \leq -\ 924 \leq -\ 726)$. The strong correspondence of this investigation is attributable to the robustness of the experimental design and the sophisticated methodology employed. Most notably, the integration of digital motion analysis via Tracker software and Fourier analysis eliminated the qualitative human observation errors in visual node identification. By tracking 19 distinct points, the design allowed for an amplitude-position calibration curve built on a fixed-frequency sinusoidal model, enabling the determination of a quantitative point of ICR.

Even though the experimental results were identical to those of the theoretical ones, their %error and the %relative uncertainty of the exponent (-1.08$\pm$0.09) and the constant (924$\pm$198), assuming its unit is cm^2, were calculated from their theoretical values of -1 and 833.33cm^2 ($\frac{L^2}{12} = \frac{100^2}{12}$) to demonstrate the limitations of the research.

	%error	%relative uncertainty
Exponent	$\left\|\frac{(-1)-(-1.083)}{-1}\right\| \times 100\% = 8.30\%$	$\left\|\frac{0.0878}{-1.083}\right\| \times 100\% = 8.11\%$
Constant	$\left\|\frac{(833.33)-(923.85)}{833.33}\right\| \times 100\% = 10.86\%$	$\left\|\frac{198}{-923.85}\right\| \times 100\% = 21.4\%$

Table 6: %error and %relative uncertainty

The numbers above indicate a noticeable deviation and a significant uncertainty, indicating both mild accuracy and low precision. To elevate the precision and accuracy, future replications of the research must address the limitations of the current apparatus.

First, while the derivation relies on a perfectly uniform moment of inertia, natural wood grain variations and the mass removed by drilling the 19 holes shifted the rod's true CM and changed its moment of inertia. Because the theoretical constants depend directly on the moment of inertia of the object, this altered mass distribution changed the rod's actual resistance to rotation, shifting the empirical constant away from its ideal value. Future replications should substitute the wooden rod with an alloy bar of uniform density and locate the true CM to integrate the true mass distribution into the theoretical model.

Second, the investigation assumes an isolated system, but sliding kinetic friction transferred mechanical energy from the oscillating rod directly into the support. This caused the support to move along with the rod, exaggerating the holes' horizontal movement and its resulting amplitude. This distorts the calibration curve used to calculate the x-intercept. This can be neutralized in future designs by clamping the framework rigidly and replacing the sliding section with a low-friction linear air track.

Third, the difference of 1mm between the 5mm drilled holes and the 4mm metal rod used to connect them to the intermediate allowed the rod to lean and wobble along the out-of-plane

z-axis. Because the Tracker measures a flat, two-dimensional projection, any twisting into the third dimension distorts the horizontal movement of the holes used to track. This warps the amplitude of the holes used to find the x-intercept. Upgrading the loose connection with flanged pivot joints would confine the system's oscillation to a two-dimensional plane.

Finally, the rod's motion blur at 60 fps visually obscured the holes at peak velocities. This twisted the tracked point from the actual motion of the holes, heavily influencing the sinusoidal model used for determining the amplitude and the ICR. Utilizing a high-speed camera at 240 fps or higher under bright, direct lighting would eliminate these motion blurs.

VII. References

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VIII. Appendix

Position of the driving point for oscillation from the CM (x_{osc}/cm) $\pm$0.05cm	**Frequency (f/Hz)**
45.00	2.669
40.00	2.669
35.00	2.742
30.00	2.669
25.00	2.675
20.00	2.669
15.00	2.806
10.00	2.608
5.00	2.602
0.00	2.669
-5.00	2.675
-10.00	2.669
-15.00	2.669
-20.00	2.669
-25.00	2.669
-30.00	2.669
-35.00	2.669
-40.00	2.669
-45.00	2.669

Appendix 1: The dominant frequency from each position along the wooden rod for the driving point 45cm away from the CM